

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 1 -- Answer Key

Q1 Solution (30 points)

a) Irreducibility:

A chain is irreducible if every state can reach every other state.

From the transition matrix, every entry is strictly positive, so in ONE step every state can reach every other state directly. Therefore the chain is irreducible.

b) Stationary distribution $\pi_i * P = \pi_i$:

The equations $\pi_i(i) = \sum_j \pi_i(j) * P(j,i)$ give:

$$\begin{aligned}\pi_i(A) &= 0.5 * \pi_i(A) + 0.1 * \pi_i(B) + 0.4 * \pi_i(C) & \dots \text{ (I)} \\ \pi_i(B) &= 0.3 * \pi_i(A) + 0.6 * \pi_i(B) + 0.1 * \pi_i(C) & \dots \text{ (II)} \\ \pi_i(C) &= 0.2 * \pi_i(A) + 0.3 * \pi_i(B) + 0.5 * \pi_i(C) & \dots \text{ (III)} \\ \pi_i(A) + \pi_i(B) + \pi_i(C) &= 1 & \dots \text{ (IV)}\end{aligned}$$

Simplify (I): $0.5 * \pi_i(A) = 0.1 * \pi_i(B) + 0.4 * \pi_i(C)$

$$5 * \pi_i(A) = \pi_i(B) + 4 * \pi_i(C) \quad \dots \text{ (I')}$$

Simplify (II): $0.4 * \pi_i(B) = 0.3 * \pi_i(A) + 0.1 * \pi_i(C)$

$$4 * \pi_i(B) = 3 * \pi_i(A) + \pi_i(C) \quad \dots \text{ (II')}$$

From (I'): $\pi_i(B) = 5 * \pi_i(A) - 4 * \pi_i(C)$

Sub into (II'): $4 * (5 * \pi_i(A) - 4 * \pi_i(C)) = 3 * \pi_i(A) + \pi_i(C)$

$$20 * \pi_i(A) - 16 * \pi_i(C) = 3 * \pi_i(A) + \pi_i(C)$$

$$17 * \pi_i(A) = 17 * \pi_i(C) \Rightarrow \pi_i(C) = \pi_i(A)$$

Then: $\pi_i(B) = 5 * \pi_i(A) - 4 * \pi_i(A) = \pi_i(A)$

From (IV): $\pi_i(A) + \pi_i(A) + \pi_i(A) = 1 \Rightarrow \pi_i(A) = 1/3$

$$\pi_i(A) = \pi_i(B) = \pi_i(C) = 1/3 \quad (\text{uniform stationary distribution})$$

c) Expected return time to A:

$$E[\text{return time to A}] = 1 / \pi_i(A) = 1 / (1/3) = 3 \text{ steps}$$

Q2 Solution (35 points)

a) Solving the hitting probability recurrence:

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 1 -- Answer Key

The characteristic equation of $h(k) = (2/3)h(k+1) + (1/3)h(k-1)$ gives:

$$r = q/p = (1/3)/(2/3) = 1/2$$

General solution: $h(k) = A + B(1/2)^k$

Apply boundary conditions:

$$h(0) = 0: A + B(1/2)^0 = 0 \Rightarrow A + B = 0 \Rightarrow B = -A$$

$$h(5) = 1: A + B(1/2)^5 = 1 \Rightarrow A - A(1/32) = 1$$

$$A(1 - 1/32) = 1 \Rightarrow A(31/32) = 1 \Rightarrow A = 32/31$$

So $B = -32/31$ and:

$$h(k) = (32/31) * [1 - (1/2)^k] = (2^5 - 2^{5-k}) / (2^5 - 1)$$

b) $h(2)$ for the biased walk:

$$h(2) = (32/31) * [1 - (1/2)^2] = (32/31) * (3/4) = 24/31 \sim 0.774$$

Symmetric walk: $h_{\text{sym}}(2) = 2/5 = 0.4$.

The upward bias ($p=2/3$) dramatically increases the probability of reaching 5 before 0: from 40% to 77.4%.

c) Symmetric walk OST -- $h_{\text{sym}}(2)$ and $E[T]$:

Martingale $S(n)$, OST (T bounded): $E[S(T)] = S(0) = 2$

$$5 * h_{\text{sym}}(2) + 0 * (1 - h_{\text{sym}}(2)) = 2 \Rightarrow h_{\text{sym}}(2) = 2/5 \quad (OK)$$

Martingale $M(n) = S(n)^2 - n$, OST: $E[M(T)] = M(0) = 4 - 0 = 4$

$$E[S(T)^2] - E[T] = 4$$

$$S(T)^2 = 25 \text{ with prob } 2/5, \quad S(T)^2 = 0 \text{ with prob } 3/5$$

$$E[S(T)^2] = 25 * (2/5) + 0 = 10$$

$$10 - E[T] = 4 \Rightarrow E[T] = 6 \text{ steps}$$

Q3 Solution (35 points)

a) Thinning property:

$$\lambda_L = \lambda * p = 4 * 0.25 = 1 \text{ latte order per hour}$$

$$N_L(t) \sim \text{Poisson}(\lambda_L * t) = \text{Poisson}(t)$$

b) P(no latte in 30 minutes):

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 1 -- Answer Key

$$P(N_L(0.5) = 0) = e^{(-\lambda_L * 0.5)} = e^{(-1 * 0.5)} = e^{(-0.5)} \sim 0.6065$$

About a 60.7% chance of no latte being ordered in the first 30 minutes.

c) Conditional distribution given 8 arrivals:

Given exactly $n = 8$ customers arrive, each independently orders a latte with probability $p = 0.25$.

The number of latte orders is:

$$\text{Binomial}(n = 8, p = 0.25)$$

Conditional expected number of latte orders:

$$E[\text{latte orders} \mid 8 \text{ arrivals}] = n * p = 8 * 0.25 = 2 \text{ orders}$$